

Deneb/Vega-Lite cheat sheet.

Quick reference for Power BI custom visuals — the blocks, the syntax, the gotchas.



Duncan Boyne
THE POWER BI KINDA GUY

01 CORE BLOCKS

THE THREE THINGS EVERY SPEC NEEDS

DATA

```
"data": { "name": "dataset" }
```

Connects Power BI data to the spec.

MARK

```
"mark": { "type": "bar" }
```

Defines the visual shape to render.

ENCODING

```
"encoding": { x, y, color }
```

Maps data fields to visual channels.

02 MARK TYPES

"MARK": { ... }

Common

bar, line, arc, circle, text, area

Other

point, rect, rule, square, tick,
geoshape

Composite

boxplot, errorband, errorbar

03 FIELD MAPPING

ENCODING "TYPE"

nominal

Category or string data.

quantitative

Numeric or measure data.

temporal

Date or datetime data. **ordinal** — ranked order for sorting.

04 SORTING & FORMATTING

ORDER & NUMBERS

Sort by axis

```
"sort": "-x" — prefix - for descending.
```

Sort by field

```
"sort": { "op": "sum", "field": "Sales" }
```

PBI format

```
"formatType": "pbiFormat" — e.g.  
"#0,,.0M".
```

05 INTERACTIVITY

PARAMS · SELECTIONS

Tooltips

```
"tooltip": true — or customise via  
encoding tooltip block.
```

Linked charts

Selection brush in chart 1, filter by param in chart 2.

Input widget

```
"bind": { "input": "range", min, max,  
step }
```

06 LAYERS & COMPOSITION

STACKING SPECS

layer

Array of specs — renders in order. **First = bottom, last = top.**

facet / concat

facet for small multiples; hconcat /
vconcat for side-by-side.

repeat

Repeats the full dataset across multiple fields.

07 COLOURS & THEMING

THEME-AWARE

Named / scheme

```
"color": "darkgreen" or "scheme":  
"category20b"
```

pbiColor

```
{ "expr": "pbiColor(0)" } — zero-based  
PBI theme index.
```

Gradients

```
"gradient": "linear" with stops, or  
"pbiColorLinear"
```

08 TRANSFORMS & PARAMS

LOGIC & REUSE

calculate

```
"calculate": "2*PI*datum['Pct']", "as":  
"Radians"
```

params

Named values and expressions reusable across the spec.

conditions

```
"condition": [{ "test", "value" }] —  
fallback via "value".
```

09 CHART SPECIFICS

TRICKS WORTH REMEMBERING

Bar

cornerRadiusEnd, stacked color encoding,
"stack": "normalize" for 100%.

Line

```
"interpolate": "cardinal", tension:  
0.7, axis format "%b %Y".
```

Arc / donut

outerRadius, innerRadius, theta / theta2
(radians, 0 = north).

